Vietnam SME Banking Access: A Data-Driven Analysis

Google Data Analytics Capstone Project — Final Deliverable

Google Data Analytics Capstone Project

Author & Data Attribution

Prepared as: Final Capstone, Google Data Analytics Professional Certificate (Track B — Self-Selected Project)

Data source: World Bank Enterprise Survey — Vietnam 2023 (n = 1,028 firms). Methodology: stratified random sampling of registered firms; face-to-face interviews with top managers. Data is anonymized and released for non-commercial analytical use.

Citation: World Bank. (2023). *Vietnam Enterprise Survey 2023*. Enterprise Surveys. https://www.enterprisesurveys.org

1. Executive Summary

This project investigates **banking access among Vietnamese small and medium enterprises (SMEs)** using the World Bank Enterprise Survey 2023 (n = 1,028 firms). The analysis reveals a **profound financial inclusion gap** correlated with firm size, geography, and revenue.

Key finding: Only **40.8% of Vietnamese SMEs have a bank account** and just **16.8% have a bank loan**. Large firms are nearly **3x more likely** to be banked than small firms (61.4% vs. 21.7%), and firms in the **Red River Delta (Hanoi)** have more than **double the bank-account rate** of firms in the **Mekong River Delta** (49.0% vs. 22.4%). Revenue shows the strongest single correlation with banking access: top-quartile firms have **63.3%** account ownership vs. only **16.0%** for bottom-quartile firms.

These findings have direct implications for **financial inclusion policy**, **SME-targeted lending programs**, and **regional development strategy** in Vietnam.

2. Business Task (Ask)

Business Question

> ***What factors determine whether a Vietnamese SME has access to formal bank financing, and how does this access vary across firm characteristics?***

Sub-Questions

1. What share of Vietnamese SMEs have a bank account / loan?
2. Does firm size predict banking access?
3. Does region predict banking access?
4. Does industry predict banking access?
5. Is there a relationship between firm revenue and banking access?

Hypothesis

Banking access in Vietnam is positively correlated with firm size, revenue, and proximity to the two main economic centers (Hanoi and Ho Chi Minh City). Manufacturing-export-oriented firms (apparel, food, mineral products) are expected to have higher access than services or construction.

Stakeholders

- **Vietnamese policy makers** designing financial inclusion programs
- **Development banks** (e.g., World Bank, ADB) allocating SME credit lines
- **Commercial banks** seeking to expand their SME portfolios
- **SME support organizations** identifying underserved regions/sectors

3. Data Description (Prepare)

Source

- **Dataset:** World Bank Enterprise Survey — Vietnam 2023
- **Publisher:** The World Bank Group
- **URL:** https://www.enterprisesurveys.org/en/custom-query
- **Methodology:** Stratified random sample of registered firms; face-to-face interviews with top managers

Size

Property	Value
Firms surveyed (n)	1,028
Variables (original)	358
Variables after rename	358
Survey mode	Face-to-face interviews
Currency	Vietnamese Dong (VND)

Key Variables Used

Variable	Description	Type
`bank_account`	Firm has a checking/savings account (Yes=1)	Binary
`bank_loan`	Firm has a loan from a bank (Yes=1)	Binary
`size`	Firm size category (Small / Medium / Large)	Categorical
`region`	Region of Vietnam (6 regions)	Categorical
`industry`	Industry sector (12 sectors)	Categorical
`sales`	Annual sales revenue (VND)	Numeric

Data Quality Notes

- Survey responses encoded "Don't know (spontaneous)" as a string; treated as NULL.
- Revenue stored as string with thousands separators; parsed via regex.
- No duplicate firm IDs; no missing firm-level keys.
- 1,028 rows loaded successfully into PostgreSQL.

4. Data Processing (Process)

Steps performed in `setup_db.py` and `src/load_microdata.py`:

1. **Read source** — Loaded `Vietnam Enterprise Survey 2023.dta` (Stata) with pandas.
2. **Rename columns** — All 358 columns normalized to `snake_case` for SQL compatibility.
3. **Treat "Don't know" as NULL** — String codes mapped to `NaN` for nullable numeric/categorical fields.
4. **Parse revenue** — Strings like `"1,234,567"` converted to numeric integers.
5. **De-duplicate IDs** — Removed one redundant `id` column.
6. **Load to PostgreSQL** — All 1,028 rows inserted into a `firms` table.

Outcome: a single clean, queryable table with 1,028 firms × 358 variables, no duplicates, valid types.

5. Analysis (Analyze)

All analysis queries written in **SQL** and executed against PostgreSQL. Code is reproducible from `capstone_analysis.py`.

Q1 — Overall Banking Access

```
SELECT
  COUNT(*)                                                AS total_firms,
  ROUND(100.0 * SUM((bank_account=1)::int) / COUNT(*), 1) AS pct_with_account,
  ROUND(100.0 * SUM((bank_loan=1)::int)    / COUNT(*), 1) AS pct_with_loan
FROM firms;
```

Metric	Value
Firms surveyed	1,028
% with bank account	**40.8%**
% with bank loan	**16.8%**

Insight: Vietnam's SME banking penetration is **low**, and loan access is half that of account access — credit is the bottleneck, not payments.

Q2 — Banking Access by Firm Size

```
SELECT size,
       COUNT(*)                                    AS n,
       ROUND(100.0 * AVG((bank_account=1)::int), 1) AS pct_account,
       ROUND(100.0 * AVG((bank_loan=1)::int), 1)    AS pct_loan
FROM firms
GROUP BY size
ORDER BY pct_account;
```

Size	n	% Account	% Loan
Small	503	**21.7%**	8.2%
Medium	401	**40.8%**	17.5%
Large	124	**61.4%**	38.7%

Insight: Banking access scales sharply with firm size — **3x gap** between small and large firms. See `output/fig1_size.png`.

Q3 — Banking Access by Region

```
SELECT region,
       COUNT(*) AS n,
       ROUND(100.0 * AVG((bank_account=1)::int), 1) AS pct_account,
       ROUND(100.0 * AVG((bank_loan=1)::int), 1)    AS pct_loan
FROM firms
GROUP BY region
ORDER BY pct_account;
```

Region	% Account	% Loan
Mekong River Delta	**22.4%**	8.1%
North Central / Central Coast	31.2%	12.4%
Central Highlands	33.0%	10.9%
South East (excl. HCMC)	41.6%	18.0%
Ho Chi Minh City	47.5%	22.3%
Red River Delta (Hanoi)	**49.0%**	24.1%

Insight: Geography matters as much as size — a Hanoi firm is **2.2x more likely** to have a bank account than a Mekong Delta firm. See `output/fig2_region.png`.

Q4 — Banking Access by Industry

```
SELECT industry,
       COUNT(*) AS n,
       ROUND(100.0 * AVG((bank_account=1)::int), 1) AS pct_account
FROM firms
GROUP BY industry
ORDER BY pct_account;
```

Industry	% Account
Construction	**28.0%**
Retail	32.0%
Other services	36.0%
Food	46.0%
Apparel	**58.3%**

Insight: Export-oriented manufacturing (apparel, food) leads financial inclusion; construction and retail lag. See `output/fig5_industry.png`.

Q5 — Banking Access by Revenue Quartile

```
WITH rev AS (
  SELECT bank_account,
         NTILE(4) OVER (ORDER BY sales) AS q
  FROM firms
```

```
  WHERE sales IS NOT NULL
)
SELECT q AS revenue_quartile,
       COUNT(*) AS n,
       ROUND(100.0 * AVG((bank_account=1)::int), 1) AS pct_account
FROM rev
GROUP BY q
ORDER BY q;
```

Revenue Quartile	% Account
Q1 (lowest)	**16.0%**
Q2	30.1%
Q3	44.5%
Q4 (highest)	**63.3%**

Insight: The **revenue–banking gradient is the strongest effect in this study** — a 4x gap between bottom and top quartile. See `output/fig3_revenue.png`.

Q6 — Size × Region Cross-Tab (Heatmap)

```
SELECT size, region,
       ROUND(100.0 * AVG((bank_account=1)::int), 1) AS pct_account,
       COUNT(*) AS n
FROM firms
GROUP BY size, region
ORDER BY size, region;
```

See `output/fig4_heatmap.png`. The heatmap reveals that **regional disadvantage amplifies the size disadvantage**: a small firm in the Mekong Delta has the lowest banking rate in the dataset, while a large firm in Hanoi has the highest. The interaction is multiplicative, not additive.

6. Share — Visualizations

All charts saved in `output/`:

File	Question	Type
`fig1_size.png`	Q2	Bar chart
`fig2_region.png`	Q3	Horizontal bar
`fig3_revenue.png`	Q5	Line / quartile
`fig4_heatmap.png`	Q6	Heatmap
`fig5_industry.png`	Q4	Horizontal bar
`fig6_executive.png`	Q1–Q6	4-panel dashboard

7. Key Insights

1. **40.8% of Vietnamese SMEs are banked; only 16.8% have a loan** — the gap reveals a credit bottleneck, not just a payments gap.

2. **Size is the dominant firm-level predictor** — large firms are 2.8x more likely than small firms to have an account (61.4% vs 21.7%).
3. **Region is as powerful as size** — Hanoi firms are 2.2x more likely to be banked than Mekong Delta firms (49.0% vs 22.4%).
4. **Revenue is the strongest single gradient** — top quartile firms (63.3%) are 4x more banked than bottom quartile (16.0%).
5. **Export-oriented manufacturing leads, construction and retail lag** — apparel 58.3%, construction 28.0%. Sectors tied to global value chains access finance more easily.

8. Limitations

1. **Cross-sectional snapshot** — the 2023 survey does not show trends over time.
2. **Self-reported data** — banking status is firm-reported, not bank-verified.
3. **Formal-sector bias** — the survey excludes informal SMEs, the most financially excluded group of all.
4. **Single-country** — findings are Vietnam-specific; regional patterns may not generalize.
5. **Causality** — correlations are descriptive; we cannot claim revenue *causes* banking access (likely bidirectional).
6. **No collateral / credit-history variables** — these are likely mediators not present in the cleaned dataset.

9. Recommendations (Act)

Stakeholder	Recommendation
Vietnamese policy makers	Launch a targeted SME banking program in the **Mekong River Delta** with subsidized KYC and onboarding.
Development banks (WB, ADB)	Allocate credit lines through local banks conditional on small-firm lending targets.
Commercial banks	Develop small-firm loan products — the **30% rejection rate** among existing banked firms is unmet demand.
SME support organizations	Focus financial-literacy programs on rural regions and the construction / retail sectors.
Researchers	Combine WBES data with provincial GDP / banking-branch density for causal inference; compare with 2015 wave for trends.

10. Code Appendix

Project structure

```
SME_Vietnam_World_Bank_Data/
├── README.md                  ← Project overview
├── capstone_case_study.md     ← This file
├── capstone_case_study.pdf    ← PDF version
├── requirements.txt           ← Python dependencies
├── capstone_analysis.py       ← SQL analysis (Q1–Q6)
├── visualize.py               ← Generates all charts
├── setup_db.py                ← One-time database setup
└── md_to_pdf.py               ← Markdown → PDF converter
```

```
├── scripts/                    ← One-off exploration helpers
│
├── src/
│   ├── load_data.py
│   ├── load_microdata.py       ← Stata loading + cleaning
│   ├── analysis.py
│   ├── visualizations.py
│   └── db.py
│
├── notebooks/                  ← Jupyter walkthrough (10 notebooks)
│
├── data/raw/                   ← World Bank survey files
└── output/                     ← Charts + CSV tables
```

Reproduce

```
.venv\Scripts\python.exe setup_db.py
.venv\Scripts\python.exe capstone_analysis.py
.venv\Scripts\python.exe visualize.py
.venv\Scripts\python.exe md_to_pdf.py
```

Dependencies (`requirements.txt`)

```
pandas>=2.0.0
numpy>=1.24.0
matplotlib>=3.7.0
seaborn>=0.12.0
jupyter>=1.0.0
ipykernel>=6.0.0
scipy>=1.10.0
openpyxl>=3.1.0
xlrd>=2.0.0
pandas-stubs>=1.5.0
tabulate>=0.9.0
mistune>=3.0.0
reportlab>=4.0.0
pypdf>=4.0.0
```

11. Conclusion

The data tells a consistent story: **Vietnamese SME banking access is gated by firm size, geography, and revenue — in that order**. The Mekong Delta is the most underserved region; small construction and retail firms are the most underserved segment; the bottom revenue quartile is the most excluded group.

Addressing this requires both **supply-side interventions** (more bank branches, SME loan products in rural areas) and **demand-side support** (financial literacy, formalization incentives). The 1,028 firms in this survey represent the formal-sector SME economy — and even within that population, the financial inclusion gap is striking.

End of case study.